

EE596H: Quiz

Total Questions: 2, Total Marks: 20, Time: 60 Mins

1. Examine whether the following optimization problem is convex problem or not? Give proper reasons.

$$\begin{aligned}
 & 2(x_1^2 + x_2^2 + 2x_1x_2) \quad \min \quad \sqrt{2x_1^2 + 3x_2^2 + x_3^2 + 4x_1x_2 + 7} + (x_1^2 + x_2^2 + x_3^2 + 1)^2 \\
 & \text{s.t.} \quad \frac{(x_1 + x_2)^2}{x_3 + 1} + x_1^8 \leq 7, \\
 & \quad \quad x_1^2 + x_2^2 + 4x_3^2 + 2x_1x_2 + 2x_1x_3 + 2x_2x_3 \leq 10, \\
 & \quad \quad x_1, x_2, x_3 \geq 0.
 \end{aligned}$$

3

7

2. Consider the problem

$$\min f(\mathbf{x}) = 7x_1 + 4x_2$$

subject to

$$3x_1 + 2x_2 \geq 5, \quad x_1 = 0 \text{ or } 1, x_2 = 0 \text{ or } 1.$$

- (a) Is the above problem convex? 2
 (b) Derive and sketch the dual function. 8
 (c) Is there any duality gap? Is Slater's condition satisfied? Justify your answer. 3

Total: 13

$$2(x_1 + x_2)^2 + x_3^2 + (\sqrt{7})^2 + x_1^2$$